

CLARA DIMARCO

Raleigh, NC | dgdimarc@ncsu.edu | [linkedin.com/in/claraleonora](https://www.linkedin.com/in/claraleonora)

EDUCATION

North Carolina State University | B.S. Computer Science, Minor in Business Administration **Expected Dec. 2026**

GPA: 3.84 | Honors: Grand Challenges Scholars Program Scholar; STARS AI Scholar

Coursework: Data Structures, Software Engineering, Software Project Management, C & Software Tools

University at Albany, SUNY | B.S. Mathematics, Magna Cum Laude

May 2022

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C, Perl, HTML/CSS

AI/ML: Large Language Models (LLMs), GPT-4, Prompt Engineering

Frameworks & Libraries: React, React Native, Next.js, FastAPI, SQLAlchemy, Tailwind CSS, Spring, Bootstrap, jQuery, Pandas

Tools & Platforms: Git, Docker, MySQL, Supabase, REST APIs, Nginx, Figma, Postman

EXPERIENCE

Software Engineer, Senior Design Capstone | AKDOT&PF Bridge Assessment Tool, NC State CCEE **Aug. 2026 – Present**

- Building a full-stack earthquake-response web application (Python, FastAPI, SQLAlchemy, React, TypeScript, MySQL, Docker, Nginx) as one of 5 engineers on a state-contracted engineering team, delivering against 10 assigned system requirements.
- Translating structural-assessment algorithms from MATLAB into Python, integrating real-time USGS earthquake data and ground-motion models to automate damage prediction across Alaska's bridge inventory.

Venture Development & Research Lead | Associated Credit Management, Hong Kong SAR **May 2026 – Jul. 2026**

- Researched online learning platforms and cross-cultural pedagogical models with an interdisciplinary team to inform engagement and retention strategy for an education-focused venture within the portfolio.
- Directed strategic business planning and formulated monetization frameworks for a 4-venture digital ecosystem, translating technical system logic (AI valuation, digital escrow) into go-to-market strategy and tiered SaaS pricing under RICS regulatory standards.

Software Engineer | NASA Jet Propulsion Laboratory, Pasadena, CA **Jun. – Aug. 2025**

- Engineered a JSON-driven web service (JavaScript, jQuery, Bootstrap) that dynamically generates configuration files, unifying multiple subsystem workflows into one scalable service for the Deep Space Network.
- Automated backend configuration workflows with Python and Perl CGI scripts, eliminating manual file generation for engineers across multiple subsystem teams.

GCSP REU Research Assistant | Game2Learn Lab, NC State University (PI: Dr. Tiffany Barnes) **May 2024 – Apr. 2025**

- Prototyped Proof-42, a cross-platform mobile app (React Native, Tailwind CSS, Supabase) gamifying propositional-logic practice with adaptive difficulty and real-time feedback.
- Developed a Disjunctive Normal Form (DNF) verification algorithm and a custom LLM prompt-engineering technique using GPT-4 that improved the reliability of AI-generated logic proofs.
- Co-authored two peer-reviewed publications in AI education and outreach.

Teaching Assistant, Introduction to Computing (OOP & Java) | NC State University **Aug. – Dec. 2024**

- Mentored 100+ undergraduates in object-oriented programming, algorithm analysis, and debugging through weekly office hours and structured code review on graded assignments.

Systems Engineer | NASA L'SPACE Mission Concept Academy **May – Aug. 2024**

- Collaborated with a cross-disciplinary, nationwide student team to develop a 199-page lunar surface mission design focused on water-ice detection, co-authoring key documentation including the Mission Definition Review, System Requirements Review, and Mission Concept Review.
- Developed scientific objectives and created a Science Traceability Matrix mapping requirements to instrumentation and performed a Failure Modes and Effects Analysis (FMEA) to identify and mitigate mission risks.

LEADERSHIP & ACTIVITIES

Treasurer & Fundraising Chair | Grand Challenges Student Organization **Aug. 2024 – Dec. 2025**

- Transitioned the club to official nonprofit status and built its financial infrastructure.

STARS AI Scholar | STARS Computing Corps **Dec. 2023 – Sep. 2025**

- Co-led AI-literacy workshops for 200+ K–12 students, including inclusive programming for students with disabilities.